

High-Dimensional Controls, Heterogeneity, and Double/Debiased ML

Empirical Methods — Week 10

Teaching deck draft

Week 10 question

- How can flexible methods help estimate causal effects without turning nuisance prediction into the answer?
- When are high-dimensional controls a design asset rather than a post-selection problem?
- How do orthogonal scores and cross-fitting protect inference?
- When is treatment-effect heterogeneity substantively meaningful?

Prediction accuracy is useful only after the estimand, timing, overlap, and identifying assumptions are explicit.

Why high-dimensional controls create new problems

- Applied data contain worker histories, firms, places, text measures, lagged outcomes, and interactions.
- Including everything can overfit, create collinearity, and weaken precision.
- Selecting controls after seeing results invalidates ordinary post-selection inference.
- Hand-picked controls can omit important confounders that predict treatment.

$$(Y_i(1), Y_i(0)) \perp D_i \mid X_i, \quad 0 < \Pr(D_i = 1 \mid X_i) < 1$$

The algorithm helps use rich observables. It does not make the observables sufficient.

Post-double selection

$$Y_i = \theta_0 D_i + g_0(X_i) + U_i, \quad D_i = m_0(X_i) + V_i$$

$$\hat{S}_Y = \text{supp}(\hat{\gamma}_Y), \quad \hat{S}_D = \text{supp}(\hat{\gamma}_D), \quad \hat{S} = \hat{S}_Y \cup \hat{S}_D$$

- Use lasso-style selection in the outcome equation and treatment equation.
- Estimate the treatment effect using the union of selected controls.
- Best when approximate sparsity is plausible and the target is low-dimensional.
- Does not solve unobserved confounding, nonlinear misspecification, or poor overlap.

Orthogonalization: DML math

$$g_0(X_i) = \mathbb{E}[Y_i \mid X_i], \quad m_0(X_i) = \mathbb{E}[D_i \mid X_i]$$

$$\tilde{Y}_i = Y_i - g_0(X_i), \quad \tilde{D}_i = D_i - m_0(X_i)$$

$$\psi(W_i; \theta, \eta) = (D_i - m(X_i)) [Y_i - g(X_i) - \theta (D_i - m(X_i))], \quad \eta = (g, m)$$

$$\hat{\theta}_{DML} = \left(\frac{1}{n} \sum_i \hat{\tilde{D}}_i^2 \right)^{-1} \left(\frac{1}{n} \sum_i \hat{\tilde{D}}_i \hat{\tilde{Y}}_i \right)$$

Orthogonality makes small nuisance errors second-order for the target score.

Cross-fitting

Split the sample into folds $I_1, \dots, I_K$. For each held-out fold:

$$\widehat{\widehat{Y}}_i = Y_i - \widehat{g}^{(-k)}(X_i), \quad \widehat{\widehat{D}}_i = D_i - \widehat{m}^{(-k)}(X_i), \quad i \in I_k$$

- Train nuisance learners on I_k^c , evaluate scores on I_k .
- Rotate across folds and stack held-out residuals.
- Tune hyperparameters inside training folds.
- Cluster folds when the identifying variation or errors are clustered.

Cross-fitting protects the target step from overfit nuisance predictions.

What DML identifies

Object		Helps when	Does not fix
High-dimensional controls	con-	many pre-treatment confounders are observed	omitted unobservables or post-treatment controls
Orthogonal score		nuisance functions are complex but auxiliary	a wrong estimand or weak design
Cross-fitting		overfit nuisance estimates threaten inference	tuning leakage or bad support
DML estimate		target is low-dimensional and design is credible	extrapolation outside overlap

Report the estimand, identifying assumption, nuisance learners, fold plan, overlap checks, and learner sensitivity.

$$\tau(x) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x]$$

$$\hat{\tau}(x) = \arg \min_{\tau} \sum_i \alpha_i(x) [(Y_i - \hat{\mu}(X_i)) - \tau (D_i - \hat{e}(X_i))]^2$$

- Forest weights $\alpha_i(x)$ define local comparison groups.
- Honesty separates split selection from effect estimation.
- Useful for heterogeneity discovery, targeting diagnostics, and mechanism hypotheses.
- Variable importance is not a causal mechanism.

Implementation pitfalls

Pitfall	Why it matters	Check
Weak overlap	flexible learners extrapolate in thin support	propensity distribution, trimming
Feature leakage	post-treatment features encode the answer	timing diagram
Learner choice	nuisance bias/variance affects target stability	lasso, ridge, forest sensitivity
Tuning leakage	full-sample tuning weakens splitting discipline	nested tuning
Fold instability	estimates depend on random split	seed and fold diagnostics
Noisy heterogeneity	CATEs overfit sparse subgroups	honest forests, hold-out validation

Reproduce

- synthetic online labor data
- baseline regression
- post-double selection
- cross-fitted DML

Diagnose

- overlap support
- nuisance R^2
- leakage audit
- fold stability

Transfer

- youth program data
- CATE by baseline risk
- honest-tree heterogeneity
- uncertainty memo

The lab reproduces design logic, not official paper estimates.

Takeaways

- Start with the economic question and target parameter.
- High-dimensional controls help only when the relevant confounders are observed and timed correctly.
- Orthogonalization protects inference from nuisance estimation error, not from bad identification.
- Cross-fitting is part of the estimator, not a cosmetic validation step.
- Causal forests organize heterogeneity, but support, uncertainty, and economics determine interpretation.
- Report these tools as research-design infrastructure, not as black-box proof.